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Path planning for autonomous mobile robots in dynamic constrained environments

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ABSTRACT

This manuscript considers an integrated path planning algorithm for autonomous mobile robots operating in dynamical constrained environments, which combines an enhanced A-star algorithm with an enhanced Dynamic Window Approach (DWA). In the global planning stage, the A* algorithm is refined through several key strategies. A redundant space mechanism is incorporated to mitigate the risk of generated paths passing too near to obstacles. Moreover, a dynamic adaptive heuristic function is developed to accelerate the node search process. The algorithm's performance is further improved by integrating a direction selection strategy alongside a method for eliminating redundant path points. For close-range obstacle avoidance and trajectory modification, enhancements are applied to DWA. The introduction of temporary target points and a deviation evaluation function effectively resolves the issue of the robot becoming trapped in local minima. Meanwhile, the weight coefficients within the comprehensive trajectory evaluation function are adaptively adjusted, leading to substantial gains in critical indicators such as total path length, computational time, and average travel speed. Simulation results verify that the proposed integrated algorithm empowers the robots to generate both feasible and reliable trajectories in complex dynamic environments, demonstrating a marked advancement in overall performance compared to baseline methods.

1. Introduction

Most of autonomous mobile robots (AMR) are typically applied in constrained spaces, including indoors, underground pipelines, and industrial warehouses [1]. Among these, AMR deployed for indoor services (see Fig. 1) not only represent a key integration of contemporary robotics technology, but also serve as a vital part of the national economy. In the context of AMR, planning a suboptimal path between the start and goal points is a pivotal technological aspect that ensures autonomy. This planning must consider both manual selection or programmed settings, and facilitating the effective navigation amidst both stationary and moving obstacles. The route created by the planning algorithm is not only related to the travelling efficiency of the robot but also directly influences the critical dimensions of task completion quality and safeness. In this paper, the term “dynamic constrained environments” mainly refers to indoor and structured constrained spaces (e.g., warehouses, corridors, underground pipelines), where the global map topology is largely stable, while the environmental dynamics primarily arise from local moving obstacles such as pedestrians, carts, or other robots. Therefore, the key challenge is to achieve safe and efficient navigation in narrow or cluttered layouts under local dynamic disturbances, rather than dealing with frequent global map changes.

Recent studies in closely related unmanned systems have further highlighted the importance of adaptive planning and collision avoidance under constrained and dynamic conditions. For example, dynamic priority guidance has been introduced to improve collision avoidance of unmanned surface vehicles (USVs) in perception-limited environments, emphasizing real-time decision adaptation under incomplete observations [2]. Moreover, energy-constrained collaborative path planning has been investigated for heterogeneous amphibious unmanned vehicles in obstacle-cluttered environments, where coordination and constraints must be balanced simultaneously [3]. In addition, learning-based

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adaptive operator selection (e.g., dueling deep Q-network) has been explored to enhance evolutionary multi-objective optimization, providing useful tools for solving complex constrained planning problems with competing objectives [4]. While these studies mainly focus on USVs or multi-vehicle collaboration, they collectively reflect a broader trend toward adaptive, constraint-aware, and robust planning mechanisms. Motivated by this trend, we focus on AMR navigation in indoor structured constrained environments and develop an adaptive hierarchical planning framework based on improved A* and improved DWA.

The application of grid maps in indoor environments is beneficial because of the inherent characteristics of these settings, which include a limited spatial extent, a known layout, and the presence of various obstacles with relatively fixed positions. The constrained nature of indoor environments is typically structured, with a limited expanse, predictable layouts, and largely fixed obstructions. These inherent characteristics make them well-suited for grid-based mapping, which efficiently discretizes space into cells to model occupancy and traversability. The implementation of grid maps enhances the efficient use of computational resources and the precise depiction of the environment. These advantages render heuristic search algorithms particularly effective in such contexts, with the A* algorithm serving as a notable example due to its efficiency and broad applicability [5]. Zhang et al. introduced an improvement to the A* algorithm specifically for addressing the path planning challenges associated with stealth drones. The original search expansion strategy was replaced with a sector expansion and bidirectional search, and the algorithm search step length was adjusted according to the threat level. This resulted in a significant improvement in search efficiency and path optimality. Concurrently, an adaptive circle optimization algorithm was employed to address the challenges posed by numerous turning points and large-angle turns in the planned path, thereby enhancing the smoothness of the UAV flight trajectory [6]. In large-scale workspaces, the computational load of the A* algorithm increases sharply with space expansion. To address this, a revised A* algorithm is formulated in Ref. [7], the key methodology which is a dual-strategy approach involving pre-computation with storage and post-path optimization. In an effort to further enhance the sleekness of paths provided by the A* algorithm, Song et al. designed three types of smoothers: a line-of-sight path smoother designed to reduce jagged lines, a refinement smoother aimed at eliminating redundant waypoints, and a triple-spline interpolation-based smoother, which significantly improves path smoothness during the motion planning of unmanned surface vessels [8]. When applied to practical scenarios, the enhancement of the A* algorithm leads to significant practical implications. For AMR in post-disaster recovery operations, achieving a path with high planning efficiency, optimality, excellent traversability, and smooth presents a critical challenge, this challenge can be tackled by a methodology that implements redundant safe space maps and constructs safe corridors. Additionally, this approach expands the obstacle detection map's scope and enhances collision detection for the ultimately planned path [9]. For outdoor environments, Erke et al. developed a heuristic function for the enhanced A* algorithm with guidelines, incorporating key points in proximity to obstacles to generate pathways that effectively circumvent them. This approach utilizes variable step lengths to minimize the algorithm's computational time [10]. By incorporating the state of goods and the adjacency matrix into the planning nodes, the approach presented in Ref. [11] formulated the container unloading task as a graph search problem, thereby achieving efficient sequencing through heuristic search process of the A* algorithm. In recent years, a growing body of research has focused on applying artificial intelligence (AI) methodologies to boost the efficacy of A* algorithm. In Ref. [12], Yang et al. employ a Generalized Voronoi Diagram to segment the ship's workspace into irregular polygons, reducing computational units, and innovatively integrates navigation features extracted from historical AIS data into the path search process, thereby enhancing key A* algorithm performance metrics like calculated time and path length. With the goal of increasing the query performance of the A algorithm [13], introduced a bidirectional search strategy. Both forward and backward lists alternately search using the current node from the opposite list as the target, while also reducing waypoints and applying smoothing treatments to the path. The self-learning capability of the algorithm is explored in Ref. [14], the Q-learning algorithm from reinforcement learning employs a dynamic strategy to refine the heuristic function, in conjunction with pruning strategies being employed to handle redundant nodes, thereby improving computational efficiency.

It is crucial to recognize that while enhancements to the A* algorithm can successfully address path planning requirements in various scenarios, the algorithm's inherent limitations prevent it from resolving dynamic obstacle avoidance problems in real time. Comparatively speaking, the Dynamic Window Approach (DWA) employs speed combinations through sampling the velocity domain of the robot to generate motion trajectory, thereby facilitating continuous dynamic obstacle avoidance.

Consequently, he research community has actively explored the application and enhancement of DWA. Lee et al. performed limited memory filtering estimation of the position, velocity, and distribution of mobile obstacles, obtaining predictive state information about the obstacles. This effectively improved the obstacle avoidance performance of existing DWA algorithms in generating trajectories [15]. Zhang et al. designed a vector field DWA algorithm for underwater robots, incorporating variable parameters and vector field histogram optimization to optimize travel path. This approach effectively addresses the issue of path deviation and improves energy efficiency [16]. Similar to A* algorithm, the DWA algorithm adjusts accordingly when implemented in actual scenarios. In Ref. [17], an energy consumption evaluation



Fig. 1. AMR operating in constrained environments.

function for the robot drive motor was incorporated, and a fuzzy logic controller was implemented, with the objective of enabling real-time tuning of the weighting factors, thus effectively ensuring the robustness and low energy consumption of the robot's operation. In the context of path generation for parking lot cleaning robots [18], addressed the issue of tire load by imposing constraints on acceleration in the DWA velocity space, thereby effectively reducing the actual error. A number of practical applications can be referred to Refs. [19] and [20]. The DWA algorithm could also be applied to local obstacle avoidance in multi-robot systems [21], enhanced it through dynamic obstacle classification, velocity evaluation, and safety distance assessment, effectively ensuring both formation maintenance and obstacle avoidance capabilities for robot teams. Of particular interest is the research on the DWA evaluation function and combination weights, which has emerged as a prominent area of focus for this algorithm. Dobrevski et al. employed a data-driven approach to coordinate the contributions of each term in the DWA's objective function adaptively. Enabling adaptive adjustments based on specific environments. This method is suitable for dynamic local planning without a global map [22]. Ovuegbe et al. adopted a similar approach to investigate the selection of optimal parameter combinations, employing Bayesian optimization to select and optimize the trajectory to reach the target point. However, due to the stringent requirements for navigation time, the trajectory became too close to the obstacles [23]. Similarly [24], optimized the speed evaluation function of DWA and added target point deviation evaluation, and employed fuzzy control rules to adjust the weights of the evaluation function. The employment of fuzzy rules has been also observed in Ref. [25]. On the whole, for the problem of local dynamic obstacle avoidance, the DWA algorithm has proven to be a robust solution. However, the algorithm possesses several limitations, including a susceptibility to local optima, high computational cost, and a lack of global planning capabilities. Furthermore, the effective mixed of the A* algorithm and the DWA offers a better solution for robotic navigation. A collaborative planning framework is constructed in Ref. [26], which enhances the global path generated by an optimized A* algorithm through incorporating an improved DWA algorithm, the proposed posture adjustment function and a fuzzy controller were upgrading the performance in dynamic environments. In coal mine tunnel environments, Gu et al. integrated target weight information into the heuristic function, ensuring a near-optimal global path while utilizing key node information for local obstacle avoidance planning, meeting robotic inspection requirements in both static and dynamic scenarios [27]. Likewise, for surface robot path planning in confined waters [28], innovatively introduces an obstacle density-dependent adjustment factor into the A* heuristic function while reducing search directions, with DWA improvements focusing on path smoothing and goal association. According to the growing variety and complexity of application environments, the integrated strategy combining global and local path planning algorithms has emerged as a central direction for developing the AMR navigation capabilities.

Inspired by the aforementioned research results, this paper focus on a path planning methodology specifically designed for dynamic constrained environments. Leveraging a division of labor that employs A* algorithm for overall route optimization and DWA algorithm for navigating dynamic local hazards, an integrated approach is formed that is highly adapted to the structured constraints of indoor environments. Firstly, a set of enhancements was proposed for the conventional A* algorithm, incorporating the introduction of redundant space configuration, a direction-based selection strategy, and a mechanism for redundant waypoint elimination. Secondly, to address dynamic obstacle avoidance, the trajectory evaluation function was refined through the integration of the DWA. Subsequently, the integration of these two components was implemented, and the proposed algorithm's performance were evaluated through simulation experiments.

The rest of this paper is arranged as follows: Section 2 contributes a brief introduction to the grid map representation and the feasible AMR kinematic models employed for algorithm development. A comprehensive description of the proposed improvements to the A* and DWA algorithms is presented in Section 3. The simulation results and corresponding analysis of these improved algorithms are discussed in Section 4. At last, Section 5 concludes the key contributions and future work.

The main contributions of this work are summarized as follows.

(1) An environment-complexity–aware improved A* algorithm is proposed by introducing an obstacle-ratio–guided adaptive heuristic and a direction-pruning node expansion strategy, which improves search efficiency while maintaining safe global paths.
(2) An improved DWA local planner is developed by incorporating temporary target points, a deviation evaluation term, and dynamic weighting coefficients, enabling robust avoidance of moving obstacles and fast return to the global reference route.
(3) A synergistic A*-DWA hierarchical planning framework is established, where the improved A* provides stable global guidance and the improved DWA performs real-time motion adaptation, achieving a better balance between global optimality and local responsiveness in dynamic constrained environments.

2. Preliminary

2.1. Grid map

Grid-based mapping serves as a simple yet efficient approach in path planning, providing an effective representation of the environment that is well-suited to the characteristics of indoor scenes. The working environment is subdivided into a series of square or rectangular areas, referred to as “grids.” Each grid is characterized by a set of binary values, indicating the presence or absence of obstacles within that specific area, thereby enabling the AMR to efficiently comprehend and respond to changes in its dynamic environment. A key strength of grid maps lies in their ability to integrate data from Lidar, cameras, and ultrasonic sensors, directly associating this information with specific grid cells. This approach thus allows for data fusion using probabilistic methods like Bayesian updating, bypassing the computationally complex steps of feature matching and data association. Moreover, this approach not only simplifies the data processing workflow but also enhances the life performance and stability of map construction. Furthermore, the precision of grid map is closely tied to the configured size of the grid cells. When computational efficiency is considered, a sufficiently small grid size can be selected to address more sophisticated data requirements. Consequently, a grid map will be formulated for indoor scenes in this paper, where white grid cells will represent obstacle-free areas accessible to AMRs, and black grid cells will indicate obstacle-filled areas that are inaccessible. Such a method offers an intuitive and easily applicable basis for planning AMR paths, significantly contributing to efficient and safe autonomous operations.

2.2. AMR kinematic modeling

To assay the efficacy of the presented algorithm, a common two wheeled differential-drive AMR was selected as the experimental platform, hereinafter referred to as “the robot” for simplicity. The robot employs a two-wheeled differential kinematic model and attains forward and turning functions by independently deciding the velocities of the left and right drive wheels. For further details, refer to Ref. [29]. The establishment of a kinematic model ensures the theoretical feasibility of the path generated by the planning algorithm. It also provides a basis for the DWA algorithm to generate and evaluate feasible control commands while predicting the future trajectory of the AMR. Therefore, a control logic that coordinates the AMR's velocity with its positional state is established. Assuming that the AMR's linear velocity at time t is v, its angular velocity is θ_t, and its heading angle is ω. If the sampling time is

Δt, then the AMR's position at time $t + \Delta t$ could be expressed as $q_{t+\Delta t} = [x_{t+\Delta t}, y_{t+\Delta t}, \theta_{t+\Delta t}]^T$, where,

$$\begin{cases} x_{t+\Delta t} = x_t + v \cdot \cos\theta_t \cdot \Delta t \\ y_{t+\Delta t} = y_t + v \cdot \sin\theta_t \cdot \Delta t \\ \theta_{t+\Delta t} = \theta_t + \omega \cdot \Delta t \end{cases} \quad (1)$$

3. Methodology

Planning in constrained environments with uneven obstacle distributions (including static and moving obstacles), this paper adopts a hierarchical strategy combining global planning and reactive local avoidance. In indoor structured scenarios, the global map topology is generally stable, and most dynamics arise from local obstacle motions; therefore, frequent incremental global replanning is unnecessary. The improved A* algorithm is used to generate a safe and efficient global reference path, while an improved DWA is employed for real-time obstacle avoidance and motion adaptation. To address the drawbacks of conventional A* (e.g., tortuous and non-smooth paths and low search efficiency), we introduce an adaptive heuristic, a direction-pruning node expansion strategy, and redundant waypoint removal. For local planning, the global route provides temporary targets for DWA, and the evaluation function is extended with a deviation term and dynamically tuned weights to enhance avoidance robustness. The overall workflow of the proposed algorithm is illustrated in Fig. 2.

3.1. Improvements to the A* algorithm

This section presents the improvements to A*, including safety inflation, adaptive heuristic design, direction-pruning expansion, and waypoint refinement.

3.1.1. Redundant safety space setting

To ensure collision-free navigation in narrow passages, obstacles in the global grid map are inflated to reserve a safety margin determined by the AMR footprint and maneuvering requirements. In the experiments, the inflation radius is set to 1.5 times the AMR footprint radius.

3.1.2. Adaptive heuristic functional

The fundamental principle underlying the A* formula is the selection of the subsequent node to be expanded, a process facilitated by an evaluation function $f(n)$. This function comprises two components: the known cost $g(n)$ from the initial node to the current node n, and the estimated cost $h(n)$ for the remaining path from n to the goal, which is typically defined as

$$f(n) = g(n) + h(n) \quad (2)$$

In the ideal scenario, when $h(n)$ is in close proximity to the factual minimum cost from node n to the target point, the A* algorithm is capable of efficiently planning the shortest path. Nonetheless, in practice, $h(n)$ should exercise caution and refrain from overestimating the real cost of reaching the target. In the event that the estimate proves to be too low, the algorithm will examine a greater number of nodes, which will consequently result in a decline in search efficiency. Unlike the conventional heuristic approaches—such as Manhattan or Euclidean distance—that form the algorithmic heritage of standard A* implementations, this paper puts forth a novel heuristic function that takes into account the merits and drawbacks of both heuristic functions.

$$h(n) = p \times \sqrt{(x_g - x_n)^2 + (y_g - y_n)^2} + (1-p)^* \left(|x_g - x_s| + |y_g - y_n| \right) \quad (3)$$

where (x_s, y_s) represents the rectangular coordinates of the initial node, (x_n, y_n) denotes the coordinates of the ongoing node, (x_g, y_g) indicates the coordinates of the target node, and p is the obstacle ratio. In Eq. (3), the heuristic term is designed as a risk-aware interpolation between two common distance metrics. The Euclidean component provides a tighter estimate in open areas, while the Manhattan component is more conservative in grid-based environments and helps avoid overly aggressive

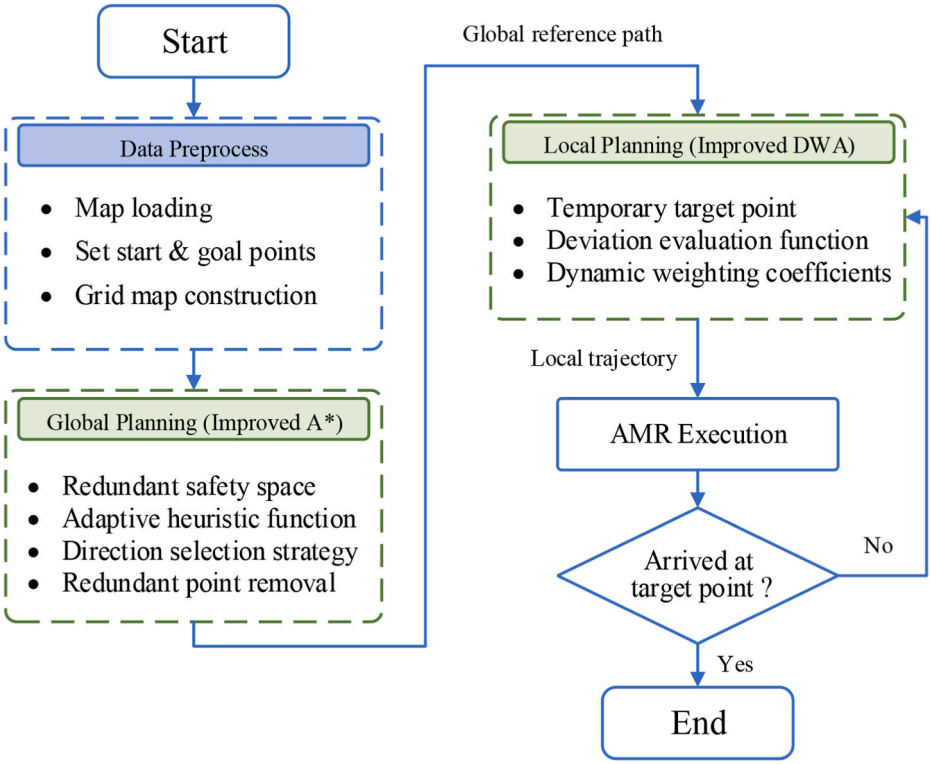


Fig. 2. The proposed algorithm flow diagram.

diagonal shortcuts near obstacles. The obstacle ratio $p \in [0,1]$ is used to smoothly shift the heuristic preference: when obstacles are dense, the heuristic becomes more conservative to encourage broader exploration; when obstacles are sparse, the heuristic remains tighter to improve search efficiency. The obstacle ratio is a key index introduced to quantify the complexity of the navigation environment. It is calculated by determining the ratio of the number of grid cells occupied by obstacles to the total number of cells within the minimal bounding rectangle defined by the start and target nodes. In the presence of a substantial number of obstacles, it is necessary to reduce $h(n)$ to expand the search range and prevent the system from becoming trapped in a local optimum. Conversely, when the number of obstacles is minimal, $h(n)$ should be increased to enhance search efficiency. On the other hand, when the weight of $h(n)$ is subject to dynamic change, particularly when the current node is distant from the target node, it is advisable to employ a larger weight. This adjustment facilitates a more expeditious expansion of the search toward the target point. In instances where the target node is found in close proximity, it is advisable to select a smaller weight to achieve an optimal path. Therefore, a dynamic adaptive heuristic function is proposed, as shown in Eq. (4), and the calculation method of weight ω is shown in Eq. (5) formula. The dynamic weight ω is introduced to adaptively regulate the influence of the heuristic term during the search process. It is scheduled according to the search progress (captured by max $\{g(n)\}$) relative to the start–goal Manhattan distance, enabling a balance between exploration and goal-directed guidance under uneven obstacle distributions. In Eq. (5), the coefficient 1.7 serves as a normalization scaling factor in an 8-connected grid map, keeping max $\{g(n)\}$ and the start–goal Manhattan distance on a comparable scale and preventing ω from becoming extreme. In practice, this design keeps ω within a moderate range and improves search efficiency while maintaining good path quality. Note that the proposed formulation focuses on improving computational efficiency; strict optimality is not explicitly claimed when $\omega > 1$, which is consistent with weighted-A style heuristic scheduling.

$$f(n) = g(n) + \omega \cdot h(n) \tag{4}$$

$$\omega = 2 - \frac{\max\{g(n)\}}{1.7 * \left\{ |x_g - x_s| + |y_g - y_s| \right\}} \tag{5}$$

3.1.3. Direction selection strategy

For the conventional A* algorithm, the parent node calculates all outlay of the eight adjacent nodes in all directions, and then chooses the node with the lowest outlay to expand as a child node. This process generates a substantial amount of computation, resulting in low search efficiency. The incorporation of a novel direction selection strategy facilitates the anticipation of obstacles, thereby reducing the number of nodes searched and enhancing search efficiency. In order to accomplish this objective, the improved algorithm requires establishing a heuristic estimate linking the current node to the destination. The angle between connecting line and positive direction of the map's x-axis is denoted by $\alpha \in [0°, 360°)$. The direction is then divided into eight equally spaced sectors based on this angle, with each sector covering an angle of 45°. In the event that α falls into the $k \in \{1,2,3,\ldots,8\}$ sector, a primary search direction is established by the center direction of that sector, in conjunction with the four expected adjacent directions. The sector number k can be calculated using Eq. (6),

$$k = \left\lfloor \frac{(\alpha + 22.5°) \bmod 360°}{45°} \right\rfloor + 1 \tag{6}$$

In addition, in the conventional A* algorithm for path planning, the heuristic function employed for node evaluation is generally based on an estimate of the straight-line distance. This results in nodes consistently expanding in the direction of the target point. However, this strategy may overlook the presence of obstacles, particularly when the obstacles are lengthy or L-shaped, thereby hindering the efficient navigation between the initial and final points. At this point, the conventional A* algorithm frequently requires a significant amount of time to bypass obstacles, thereby reducing search performance. In the designed algorithm logic, the direction of obstacle detection is based on the direction of the line connecting the current node to the target point. The current node is designated as the center, where the effective detection range is characterized by a radial extent. This configuration is utilized to ascertain the presence of obstacles in the five directions within the specified range. In the event that an obstruction is encountered in a specific direction, it is necessary to eliminate that direction and designate the adjacent direction node as a replacement. Finally, it is imperative to prioritize nodes with lower estimates for expansion. The implementation of this direction selection strategy enables the algorithm to circumvent obstacles from the starting position to the navigational objective, thereby minimizing unnecessary node expansion. Therefore, Fig. 3(a) depicts the searching scheme for the eight neighboring nodes surrounding the current node. Meanwhile, Fig. 3(b) demonstrates the path search directions employed by the A* algorithm utilizing the proposed direction selection strategy, when the angle α is located in sector 3.

3.1.4. Redundant points removal

The conventional A* algorithm is constrained in its ability to expand to eight adjacent directions when expanding nodes. Consequently, the path that can be planned is limited to the horizontal, vertical, or 45° diagonal directions. This restriction leads to the generation of unnecessary turning points and elongated paths. A method is proposed for the purpose of optimizing paths, with the objective of eliminating redundant turning points as well as reducing road length. The redundant point deletion mechanism operates through a sequential two-step process:

Step 1: The initial step involves a directional coherence check to assess path linearity. This is done by determining whether the expansion trajectory of the parent node (relative to the current node) aligns with the direction established by the previous node, with this determination being made from the target node. Conversely, if the straight path hypothesis is valid, it could be deduced that the two nodes in question belong to the same straight path, thereby eliminating the possibility of inflection points. In the event that the aforementioned condition is not met, the current node constitutes an inflection point on the complete path. According to the aforementioned judgment rules, it is necessary to extract all inflection points on the complete path.

Step 2: The starting node is designated as I_1, the endpoint is denoted by I_n, and $I_2, I_3, \ldots, I_{n-1}$ represents all the turning points in between. From the starting point I_n, a line is constructed that extends to I_1. The presence of any obstacles along this line is then determined. In the event that this is not feasible, the next step would be to delete the other redundant turning points in between. In the event that this is the case, the connection between I_n and turning point I_2 should be established, and the determination should be pursued until turning point I_{n-2} is

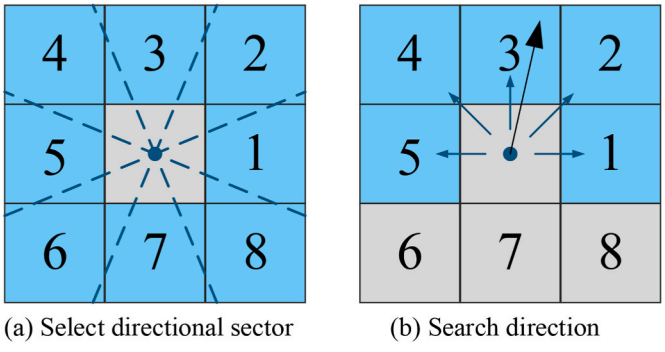


Fig. 3. Direction selection strategy.

reached. Subsequently, commencing from the designated turning point I_{n-1}, establish the connection between points I_1 and $I_2, I_3, \ldots, I_{n-3}$ in the prescribed sequence. Thereafter, undertake the requisite calculations to arrive at the desired conclusion. It is imperative to reiterate the aforementioned steps until reaching point I_3, thereby finalizing the optimization process.

As manifested in Fig. 4, the black line represents the original path. Once the redundant point removal process has been performed, the red points would be eliminated, thereupon blue path points are retained. The final blue line demonstrates a substantial improvement over the original path.

3.2. Improvements to the DWA algorithm

This subsection presents three improvements to DWA: temporary target guidance, a deviation evaluation term, and dynamically adaptive weights.

3.2.1. Core parameter configuration of the DWA algorithm

The DWA is a widely used algorithm for local motion control and obstacle interaction module in mobile robots. The fundamental principle of the DWA method involves sampling the velocity space to generate multiple feasible motion commands. The efficacy of each command is assessed by predicting the corresponding trajectory over a specified future period and evaluating it using a multi-criteria evaluation function. The command that generates the highest-scoring trajectory is selected for execution, and this process iterates at each time step [30]. The fact that a robot's movement trajectory is constrained by its own structural design and the external environment necessitates the implementation of well-defined speed limit V_m.

$$V_m = \left\{ \begin{array}{l} (v,\omega) | v \in [v_{min}, v_{max}] \\ (v,\omega) | \omega \in [\omega_{min}, \omega_{max}] \end{array} \right\} \tag{7}$$

where, v_{min} and v_{max} represent the minimum and maximum linear velocities of the robot, respectively, while ω_{min} and ω_{max} denote the minimum and maximum angular velocities of the robot. Additionally, there are constraints V_d on the drive acceleration.

$$V_d = \left\{ \begin{array}{l} (v,\omega) | v \in [v_c - \dot{v}_b \cdot \Delta t, v_c + \dot{v}_a \cdot \Delta t] \\ (v,\omega) | \omega \in [\omega_c - \dot{\omega}_b \cdot \Delta t, \omega_c + \dot{\omega}_a \cdot \Delta t] \end{array} \right\} \tag{8}$$

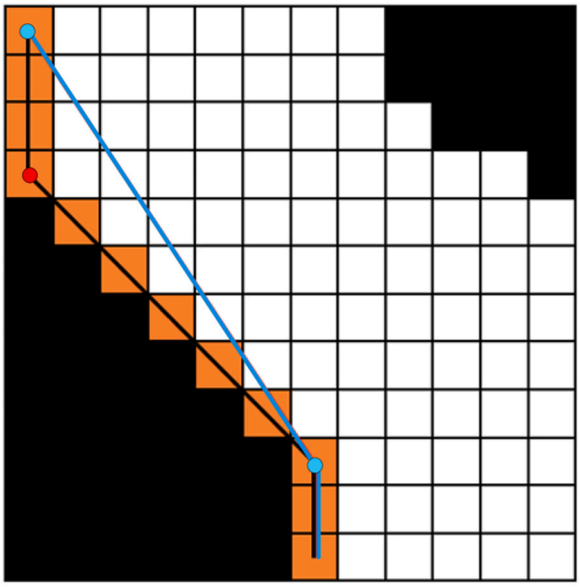


Fig. 4. Redundant points removal.

where, v_c and ω_c represent the linear velocity and angular velocity of the robot at the current time, respectively. Then, $\dot{v}_a$ and $\dot{v}_b$ denote the robot's maximum acceleration and maximum deceleration, while $\dot{\omega}_a$ and $\dot{\omega}_b$ represent its maximum angular acceleration and maximum angular deceleration, respectively.

Furthermore, when implementing dynamic obstacle avoidance for robots, it is imperative to consider the robot's motion inertia. Consequently, the speed sampling space must also take into account the robot's braking distance constraints V_a:

$$V_a = \left\{ \begin{array}{l} (v,\omega) \Big| v \le \sqrt{2 \cdot dist(v,\omega) \cdot \dot{v}_b} \\ (v,\omega) \Big| \omega \le \sqrt{2 \cdot dist(v,\omega) \cdot \dot{\omega}_b} \end{array} \right\} \tag{9}$$

among them, $dist(v,\omega)$ is the shortest distance between the simulated trajectory corresponding to (v,ω) and the obstacle. In light of the aforementioned constraints, the speed sampling space V_s of the robot is determined as follows:

$$V_s = V_m \cap V_d \cap V_a \tag{10}$$

To efficiently select the optimal trajectory in the speed sampling space, a trajectory evaluation function was constructed based on the robot kinematic model. This function selects a trajectory that best meets the robot's motion requirements by evaluating the direction, distance, and speed of each trajectory. The trajectory evaluation function is defined as follows:

$$G(v,\omega) = \sigma \cdot (\alpha \cdot heading(v,\omega) + \beta \cdot dist(v,\omega) + \gamma \cdot vel(v,\omega)) \tag{11}$$

where, $G(v,\omega)$ is the trajectory evaluation function, $heading(v,\omega)$, $dist(v,\omega)$, and $vel(v,\omega)$ are the evaluation functions for direction, distance, and speed, respectively, α, β, and γ are the weighting coefficients for each evaluation item, and σ is the normalization coefficient used to unify the units of measurement for each evaluation item.

3.2.2. Temporary target point

Through the configurations specified in Section 3.2.1, the DWA algorithm is able to empower the AMR to achieve the optimal speed in the current cycle during local planning. Nevertheless, the algorithm remains susceptible to local optimal solutions in environments characterized by complexity. Given the robot's capacity to formulate pathways solely on the basis of incomplete data derived from local perception, there is a possibility that it may become entrapped within a local optimal solution, thereby hindering its ability to successfully identify the global optimal path. To address this limitation, a dual approach of global and local path planning is employed. Within the integrated planning architecture, the global path generated by the A* algorithm is first analyzed to extract a sequence of strategic waypoints. These waypoints are then sequentially assigned as intermediate goals to the local planner. This process ensures a coherent transition from global planning to local execution, where the DWA navigates toward each successive target as the AMR progresses, insuring that the robot always moves toward the global target node during local planning. This methodology enables the robot to circumvent dynamic obstacles and mitigate path planning errors attributable to inadequate local information, thereby facilitating more efficient and intelligent autonomous navigation.

3.2.3. Deviation evaluation function

In order to facilitate the AMR's swift return to the global optimal path following successful avoidance of obstacles, the concept of deviation distance is incorporated into the DWA algorithm. Specifically, a deviation evaluation function, designated as $dev(v,\omega)$, is integrated into the path evaluation function to assess the extent of deviation between the current trajectory and the global path. This evaluation function is mathematically expressed as follows:

$$dev(v,\omega) = 20 \times \frac{1}{0.5 + d} \tag{12}$$

where, d represents the cross-track error between the robot's instantaneous path and its predefined global path. When d is large, it indicates that the robot has deviated far from the global path, and the deviation evaluation function score will decrease accordingly. Through real-time assessment of the degree of deviation, the robot can be effectively guided to quickly resume following the nominal path upon successful navigation around dynamic obstacles, thereby reducing the additional travel and time consumption caused by deviation.

3.2.4. Dynamic weighting coefficient

For the DWA algorithm, the weight coefficients of each evaluation function have a significant impact on the performance of the algorithm. The conventional DWA method generally employs fixed weight coefficients. While this design demonstrates efficacy in certain scenarios, fixed weights frequently prove ineffective when adapting to the variability inherent in dynamic constraint environments. This deficiency often leads to a decline in obstacle avoidance success rates or the emergence of local optima. Therefore, parameter tuning and adaptive coefficient design for the DWA evaluation function have been widely investigated to improve robustness and adaptability, especially in dynamic or obstacle-cluttered scenarios [31]. [32]. Consequently, the weight coefficients of the trajectory evaluation functions undergo dynamic enhancement. This method dynamically regulates the weights of specific evaluation function based on the relative positions between the robot and environmental elements, such as target points and obstacles. This enables the algorithm to adapt more flexibly to the environment, thereby improving obstacle avoidance success rates and planning effectiveness. The improved weight coefficient calculation formulas are shown in Eq. (13)–(15). It should be noted that the coefficients are designed to vary with normalized task progress and obstacle proximity, which reflect typical kinematic and safety requirements of AMRs. In Eqs. (13) and (15), the constant 1.2 is introduced as a normalization buffer to smooth the exponential transition and prevent abrupt weight changes when the robot approaches the target, thereby reducing oscillations under discretized velocity sampling and bounded acceleration/turning limits. The exponential forms for α and γ are selected to provide smooth and monotonic scheduling between goal-directed motion and stable convergence, while the logarithmic form $\ln(12 - dist)$ for β , increases sensitivity in close-range obstacle interactions. Here, 12 represents the effective local obstacle-interaction distance (consistent with the local planning horizon in the simulation), ensuring β remains well-defined and can be scaled according to robot size and safety margin for different platforms. Based on the above adaptive coefficients, the final improved trajectory evaluation function is defined as Eq. (16).

$$\alpha = 2 \times e^{1 - \frac{length(P_c, P_t)}{1.2 \times length(P_s, P_t)}} \tag{13}$$

$$\beta = \ln(12 - \mathrm{dist}) \tag{14}$$

$$\gamma = e^{\frac{length(P_c, P_t)}{1.2 \times length(P_s, P_t)}} \tag{15}$$

$$G(v,\omega) = \sigma \cdot (\alpha \cdot (heading(v,\omega) + dev(v,\omega)) + \beta \cdot dist(v,\omega) + \gamma \cdot vel(v,\omega)) \tag{16}$$

where, $P_s = (x_s, y_s)$ serves as the starting point coordinate, $P_t = (x_t, y_t)$ denotes the current target point coordinate, $P_c = (x_c, y_c)$ indicates the current point coordinate, $length(P_c, P_t)$ is the Euclidean distance between the current and target points, and $length(P_s, P_t)$ is the Euclidean distance between the starting or previous temporary target point and the current target point. The adjustment rules for the dynamic weighting coefficients are specified as follows:

1) Adjustment of the direction angle and deviation weight coefficient α. When the AMR is at a considerable distance from the target point, the value of α is small. This allows the robot to deviate slightly from the global path during local obstacle avoidance. As the robot approaches the target point, the value of α progressively increases, guiding the robot to follow the global optimal path more closely. The value range of α is $[2, 2e]$.
2) Adjustment of the distance weighting coefficient β. While the AMR runs in a remote zone relative to the obstacle's immediate vicinity, the value of β is significant, and the weight of the distance evaluation function is minimal. As the robot gradually approaches the obstacle, the value of β decreases, thereby increasing the weight of the distance evaluation function. This, in turn, compels the AMR to execute a collision avoidance maneuver by increasing its distance from the obstacle. The value range of β is $[\ln 2, \ln 9]$.
3) Adjustment of the speed weighting coefficient γ. While the AMR is located in a remote operational zone relative to the target point, the value of γ is large, enabling the robot to move quickly toward the target point. As the robot approaches the target point, the value of γ decreases in order to facilitate stable convergence toward the target by mitigating the risk of an uncontrolled approach due to excessive momentum. The range of values for γ is specified as $[1, e]$.

The determination of the aforementioned dynamic parameter ranges was informed by both domain-specific literature and iterative simulation experiments. A critical factor motivating this tailored approach is the need to accommodate the different kinematic and dynamic constraints inherent to different AMR platforms, which is essential for the effective practical application of the theoretical algorithm. In the light of the content of this part, the complete process of the proposed (Improved A*-DWA) algorithm could be obtained, as demonstrated in Table 1.

4. Simulation results and discussions

To evaluate the effectiveness of the proposed path planning strategy in constrained environments with static and dynamic obstacles, a series of simulation experiments were conducted from both global and local perspectives. The global planner was tested on different indoor grid maps with multiple start–goal configurations, while the local planner was validated under static, dynamic, and mixed obstacle settings. The integrated A*-DWA framework was further examined to demonstrate its capability of balancing global guidance and real-time obstacle avoidance. Unless otherwise stated, the simulation parameters follow the settings in Table 2.

4.1. Performance evaluation of the improved A* algorithm

The improved A* algorithm was evaluated in two representative indoor structured environments, denoted as Environment 1 and Environment 2. Each environment contains typical constrained structures such as narrow passages and uneven obstacle distributions. To ensure a comprehensive assessment, three start–goal pairs were designed in each environment, resulting in different navigation difficulties in terms of obstacle density and geometric constraints. The detailed configurations of these planning tasks are summarized in Table 3. And the specific setups are shown in Fig. 5.

4.1.1. Ablation-style comparison of improvement modules

To clarify the contribution of each improvement component in the global planner, four variants were considered: A*: conventional A* algorithm; A1: A* with adaptive heuristic weighting; A2: A1 combined with the direction selection strategy; A3: A2 further integrated with redundant waypoint removal. For each test case listed in Table 3, the four variants were executed under identical simulation settings. The performance was evaluated using the path length, the number of turning points, and the computation time. The quantitative results are reported

Table 1
Pseudocode of the proposed algorithm.

Improved A*-DWA algorithm
Initialization:
Construct a global grid map.
Initialize OPEN LIST and CLOSE LIST.
The starting node is to be appended to the open list.
Initialize the robot's current position and speed.
while the robot has not reached the target position **do**
if the open list is empty **then**
Terminate the A* algorithm, planning failed
else
Select the node with smallest evaluation function value from the open list as the current node.
if the current node is the target node **then**
Stopping the A* algorithm and generate the global path.
else
Use the direction selection strategy to expand the child nodes of the current node
Calculate the evaluation function of the child nodes based on environmental information
Add child nodes that are not in the closed list to the open list
Add the current node to the closed list
end if
end if
end while
Delete redundant nodes from the global path
while the robot did not reach the target point **do**
Sampling the robot speed
Selecting the optimal speed pair based on the evaluation function of the DWA algorithm
if the robot has reached the current temporary target point **then**
Update the next temporary target point
end if
if the robot has reached its final destination **then**
Stopping DWA algorithm
else
Execute the selected optimal speed pair and continue movement.
end if
end while
Return robot motion path

Table 2
Simulation environment parameter settings.

Simulation parameters	Settings
Map size	52m × 35m
Grid cell	0.5m
Grid expansion size	0.75m
Maximum speed	2 m/s
Maximum angular velocity	20 rad/s
Maximum acceleration	0.2 m/s^2
Maximum angular acceleration	50 rad/s^2
Speed resolution	0.01 m/s
Angular velocity resolution	1 rad/s
Sampling time	3s
Dynamic obstacle radius	0.5m

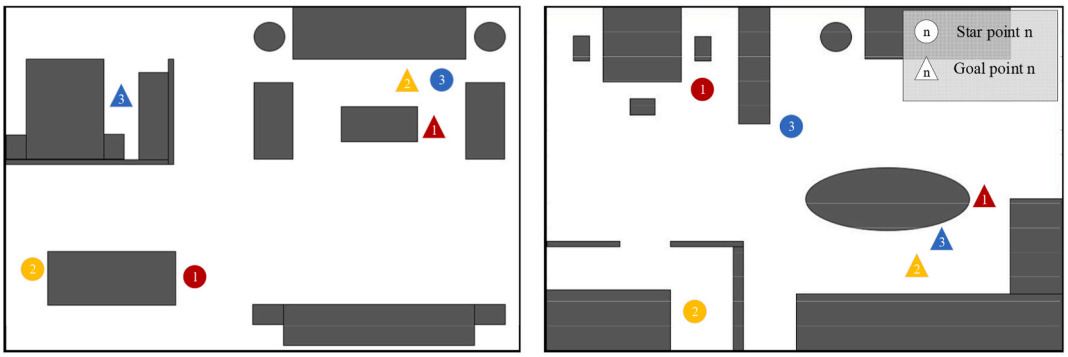
Table 3
Start–goal configurations designed in Environment 1 and Environment 2.

Environment	Case	Start point (x, y) (m)	Goal point (x, y) (m)
Environment 1	Case 1	(20,7.5)	(44,54)
	Case 2	(2.5,7.5)	(42.5,28)
	Case 3	(44.5,27)	(12,26)
Environment 2	Case 1	(16,26)	(45.5,17)
	Case 2	(15,4.5)	(38.5,9)
	Case 3	(25,24)	(40.5,10.5)

in Table 4.

As shown in Table 4, the adaptive heuristic weighting in A1 reduces unnecessary node expansions in low-complexity regions, leading to improved search efficiency compared with conventional A*. With the direction selection strategy enabled in A2, the computation time is further reduced, indicating that the node expansion process becomes more targeted. In addition, the redundant waypoint removal applied in A3 effectively decreases the number of turning points, producing a smoother global reference path which is beneficial for subsequent local trajectory tracking. Overall, the proposed improvements provide consistent gains in both planning efficiency and path quality across different environments and start–goal configurations. To more clearly demonstrate the advantages of the improved A* algorithm, we select Case 2 in the typical Environment 1 as an example for further analysis. Initially, the environment was discretized into a grid map, and static obstacles were labeled for the sake of convenience in discussion. The simulation results of global path planning are displayed in Fig. 6. The colored grids depicted in the figure represent the areas explored by the algorithm during the process of path planning. The colors of the grids range from blue to red, with the color sequence denoting the order in which the exploration areas were visited. The red curve signifies the final generated path.

In the experiment, the target point was set to the right of the starting point (next to obstacle ⑥). As illustrated in Fig. 5(a), due to the obstruction of obstacle ⑥ in the environment, the robot must initially traverse its left side before proceeding rightward to arrive at the target point. The conventional A* method, when searching for a path, typically moves in a straight line during the initial stage due to the guidance of the heuristic function. However, when confronted with impediments, the search algorithm expands to an excessive number of superfluous nodes, leading to a substantial expenditure of time to circumvent these obstacles, resulting in an augmented search path. In the enhanced A* algorithm experiment, as depicted in Fig. 5(b), the direction selection strategy and the adaptive heuristic function facilitate the effective reduction in the number of search grids, thereby enhancing search efficiency and enabling a more expeditious identification of detour paths. Moreover, following the elimination of superfluous nodes, as opposed to the conventional A* algorithm, the path length is decreased by 10.96%, and the search time is reduced by 52.81%



(a) Setup of environment1. (b) Setup of environment2.

Fig. 5. Schematic diagram of Environment 1 and Environment 2 used in the experiments.

Table 4
Performance comparison of different A* variants in global path planning.

Environment	Case	Method	Path length (m)	Number of turning points	Computation time (s)
Environment 1	Case 1	A*	33.59	3	3.12
		A1	33.59	3	2.98
		A2	36.34	9	3.17
		A3 (Improved A*)	31.33	1	3.42
	Case 2	A*	55.91	10	12.08
		A1	54.63	10	11.05
		A2	53.96	10	6.10
		A3 (Improved A*)	49.78	4	5.70
	Case 3	A*	60.02	9	35.68
		A1	60.02	15	18.52
		A2	60.79	11	23.53
		A3 (Improved A*)	58.81	7	23.58
Environment 2	Case 1	A*	37.00	9	17.32
		A1	39.21	19	8.45
		A2	38.11	9	10.92
		A3 (Improved A*)	35.74	2	10.90
	Case 2	A*	41.19	6	36.16
		A1	41.19	6	33.64
		A2	41.19	6	22.17
		A3 (Improved A*)	39.87	5	22.53
	Case 3	A*	29.18	4	38.78
		A1	36.45	23	8.97
		A2	76.65	9	13.92
		A3 (Improved A*)	74.61	3	13.55

4.1.2. Comparison with other global planning algorithms

To further assess the competitiveness of the proposed global planner, two commonly used baseline methods were included for comparison, namely Rapidly-exploring Random Tree (RRT) and Ant Colony Optimization (ACO). These algorithms represent typical sampling-based and population-based global planning strategies, respectively. Under the same test cases in Table 3, the planned paths were generated and compared in terms of path length and computation time. The results are summarized in Table 5.

According to Table 5, the improved A* algorithm exhibits stable performance in constrained indoor environments. Compared with RRT, the proposed method produces more structured paths with fewer detours in corridor-like regions. Compared with ACO, the improved A* demonstrates higher computational efficiency while maintaining comparable path quality. These results indicate that the improved A* planner provides an effective global solution for indoor constrained navigation problems.

4.2. Improved DWA algorithm for local path planning

The DWA algorithm is particularly suited for path planning in environments with changing conditions. However, it is susceptible to suffering from premature convergence, a phenomenon that is particularly evident in environments replete with obstacles. This can result in the robot traversing a limited spatial domain and failing to identify the path which is better than others. To substantiate the efficacy of the proposed dynamic weighting coefficients and assess the algorithm's dynamic obstacle avoidance capabilities, two dynamic obstacles (blue spherical objects) moving at a constant speed were incorporated into the original environment for experimentation, as show in Fig. 7.

As shown in Fig. 6(a), the robot begins at the initial point and moves gradually toward the target point under the influence of the direction evaluation function. In the event of encountering a dynamic obstacle, the robot is capable of avoiding it in a timely manner, thereby enabling the continuation of movement toward the target point. However, when the robot's motion approaches the static obstacle ③ illustrated in the figure, a typical local optimum phenomenon occurs. In other words, as the robot approaches the obstacle while moving toward the target point, its actions become characterized by decreased speed and increased rigidity. At this juncture, the robot is incapable of effectively circumventing the obstacle or persisting in its forward movement over a brief interval. Following an extended interval of gradual movement, the robot successfully navigates an obstacle and arrives at the designated final target point after traversing the constricted passageway.

Consequently, the designed deviation function and dynamic weighting coefficients were introduced, and the experiment was conducted once more. As shown in Fig. 6(b), at the beginning of the experiment, when the robot started from the initial point, due to the presence of a static obstacle ⑥ on its right side, the weight of the distance evaluation function was relatively large. Therefore, the robot chose to drive to the left to avoid the obstacle on the right. This indicates that when approaching an obstacle, the algorithm could sensitively identify it and assign a greater weight to the distance evaluation function, thereby guiding the robot to avoid obstacle. As the robot traverses a specific distance, the weights of the deviation evaluation function and the speed evaluation function undergo a gradual increase. This facilitates the robot's ability to swiftly modify its trajectory and expedite movement toward the designated target point. The actions undertaken during this phase illustrate the efficacy of the proposed algorithm in prioritizing the optimization of speed and direction while ensuring safety. This capability enables the robot to swiftly approach the target

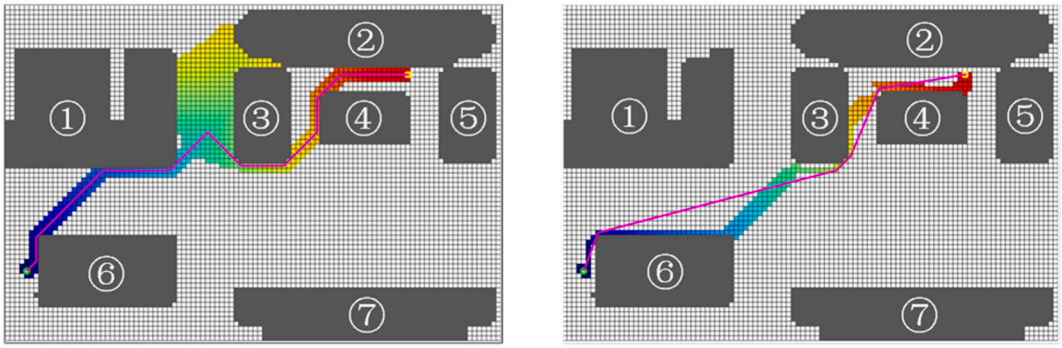


Fig. 6. Performance comparison of the A* algorithm before and after improvement.

Table 5
Comparison between the improved A* algorithm and other global planning baselines.

Environment	Case	Method	Path length (m)	Computation time (s)
Environment 1	Case 1	Improved A*	31.33	8.42
		RRT	41.41	9.76
		ACO	50.99	940.91
	Case 2	Improved A*	49.78	12.05
		RRT	71.53	10.39
		ACO	127.09	946.32
	Case 3	Improved A*	58.81	63.58
		RRT	76.32	30.59
		ACO	101.78	940.75
Environment 2	Case 1	Improved A*	35.73	10.9
		RRT	54.13	7.97
		ACO	88.71	942.01
	Case 2	Improved A*	39.87	22.53
		RRT	86.29	12.29
		ACO	101.65	943.48
	Case 3	Improved A*	32.76	13.55
		RRT	41.46	9.41
		ACO	77.42	940.56

point. However, upon the robot's approach to a dynamic obstacle, the weight of the distance evaluation function undergoes an increase, enabling the robot to promptly adjust its heading and successfully evade the dynamic obstacle. Following the successful avoidance of the obstacle, the robot swiftly resumes its forward movement and continues moving toward the target point until it finally arrives. The results of path planning using the DWA algorithm before and after improvement are compared in Table 6.

The experimental results demonstrate that the introduction of dynamic weight coefficients results in robot motion that is smoother and more efficient, particularly when dealing with dynamic obstacles. This enables rapid path adjustment to avoid getting stuck in local optima. This enhancement has been demonstrated to improve the robot's capacity for navigating obstacles and attaining objectives in complex environments. The incorporation of dynamic weight coefficients has been shown to enhance the flexibility and robustness of the DWA algorithm when confronted with dynamic and static obstacles. This augmentation empowers the robot to adapt more intelligently to complex and ever-changing environments.

Table 6
Comparison of results between the DWA algorithms.

Algorithm	Path length (m)	Planning period (s)	Average speed (m/s)
The conventional DWA	65.71	124.15	0.53
The improved DWA	68.27	75.23	0.91

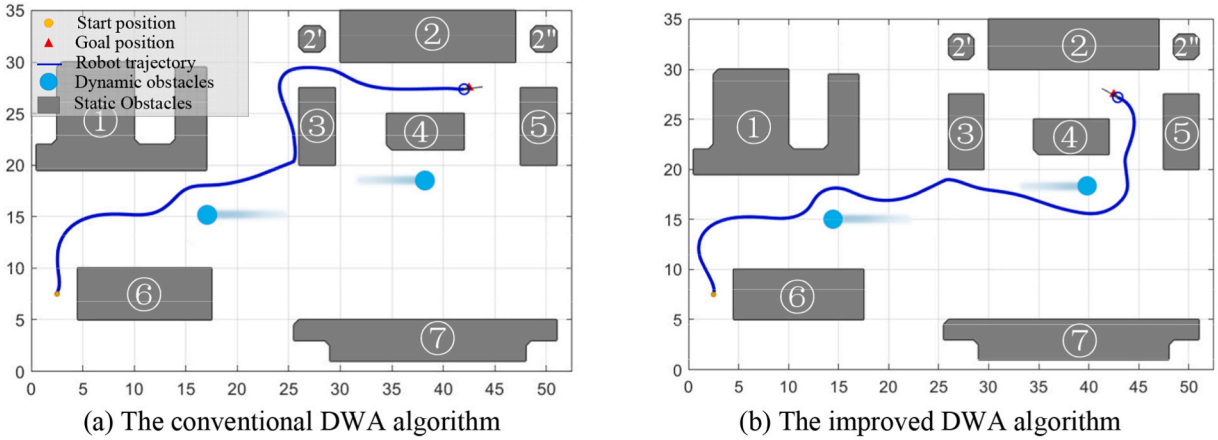


Fig. 7. Performance comparison of the DWA algorithm before and after improvement. Gray blocks labeled ①–⑦ denote static obstacles. Cyan circles denote dynamic obstacles moving at a constant speed. The yellow point is the start position and the red triangle indicates the final goal. The blue curve represents the planned robot trajectory.

4.3. A*-DWA algorithm mixed path planning

While the improved DWA algorithm can to some extent mitigate the local optimality issue, the scope of local path planning remains constrained. Therefore, the A-DWA mixed planning strategy is further developed by incorporating the global path inflection points generated by the improved A algorithm as temporary target points for local planning, together with the deviation evaluation term to encourage fast return to the global reference route. To evaluate the effectiveness of the proposed A*-DWA framework under different local obstacle configurations, three experiments were conducted in Environment 1, including: Experiment I (static obstacles): two static obstacles were added into the workspace; Experiment II (mixed obstacles): one static obstacle and one dynamic obstacle were introduced simultaneously; Experiment III (dynamic obstacles): two dynamic obstacles moving at constant velocity were introduced. The detailed experimental configurations are summarized in Table 7, and the quantitative comparison results between the improved DWA and the A*-DWA framework are reported in Table 8.

As indicated in Table 8, the proposed A*-DWA framework consistently reduces the planning period and improves the trajectory efficiency compared with the improved DWA alone, especially in scenarios involving dynamic obstacle interactions. To provide a more intuitive understanding of the underlying planning behavior, Experiment III is selected as a representative case for qualitative analysis, and the corresponding trajectory evolution is illustrated in Fig. 8.

As shown in Fig. 8, after the robot commences movement from initial node, its proximity to obstacle ⑥ results in a substantial weight attributed to the distance evaluation function. The robot's movement is influenced by this evaluation function, causing it to move away from the obstacle. On the grounds that constraints imposed by the direction evaluation function and the deviation evaluation function, the robot does not adopt the conservative obstacle avoidance strategy depicted in Fig. 6(b). Instead, it gradually approaches and reaches the temporary target point 1. Subsequently, the robot continues to move toward the temporary target point 2, successfully avoiding the dynamic obstacle, and then quickly returns to the path planned by the improved A* algorithm. The robot's trajectory is influenced by the presence of temporary target points, leading it to bypass the space below obstacle ④. Constrained by the deviation and direction functions, it selects the space between obstacles ③, ④, and ②, which can be traversed to reach the final target point 5. The robot's runtime and path length are compared with those of the improved DWA algorithm alone, as shown in Table 8.

The results demonstrate that the proposed A-DWA framework improves both planning efficiency and trajectory stability by combining global guidance from the improved A algorithm with reactive local adaptation. This integration allows the robot to avoid static and dynamic obstacles while maintaining better consistency with the global reference route in constrained environments.

5. Conclusions

In dynamic constrained environments, especially indoor scenes, the intelligent control of autonomous mobile robots has become a key research focus and development direction. Robust path planning capability is indispensable among its essential functions. In the interest of address the challenges associated with finding the globally optimal path, dynamic obstacle avoidance, and avoiding local optima, a fusion path planning methodology is introduced in this work, merging an augmented A* global planner with a refined DWA local planner. The conventional A* algorithm has been improved through the introduction of techniques such as redundant safety space definition, adaptive heuristic functions, direction selection strategies, and redundant point deletion. These techniques have been shown to significantly enhance the efficiency of global path planning and the smoothness of the paths. Motivated by the shortcomings of the conventional DWA, such as getting stuck in local optima and lacking global planning capabilities, temporary target points and deviation evaluation functions were added, and dynamic weight coefficients were adopted for the overall trajectory evaluation function. The findings of the simulation experiments have demonstrated that the enhanced A* algorithm has the capacity to reduce path length by 10.96% and planning time by 52.81%. The advanced DWA algorithm has been shown to augment the AMR's capacity for obstacle avoidance and enhance its planning efficiency in dynamic environments, resulting in an average speed increase of 41.76%. The hybrid algorithm, which integrates A* and DWA, has been observed to

Table 7
Experimental configurations for evaluating the A*-DWA mixed planning framework in Environment 1.

Parameter	Experiment I	Experiment II	Experiment III
Start point (x, y) (m)	(20, 7.5)	(2.5,7.5)	(2.5,7.5)
Goal point (x, y) (m)	(44, 22.5)	(42.5,28)	(42.5,28)
Obstacle 1 position (x, y) (m)	(12.5, 5)	(12.5, 15)	(·, 30)
Obstacle 1 velocity (m/s)	0	0	0.7
Obstacle 2 position (x, y) (m)	(37.5, 19.5)	(20, ·)	(20, ·)
Obstacle 2 velocity (m/s)	0	0.7	0.7

Table 8
Quantitative comparison results of local planning performance in Experiments I–III.

Experiment	Method	Path length (m)	Planning period (s)	Average speed (m/s)	Reduction of time (%)
I	The improved DWA	52.38	56.30	0.93	-
I	The A*-DWA	38.34	37.51	1.02	26.29
II	The improved DWA	-	(fail)	-	-
II	The A*-DWA	57.56	51.85	1.11	-
III	The improved DWA	68.27	75.23	0.91	-
III	The A*-DWA	59.71	57.87	1.03	23.1

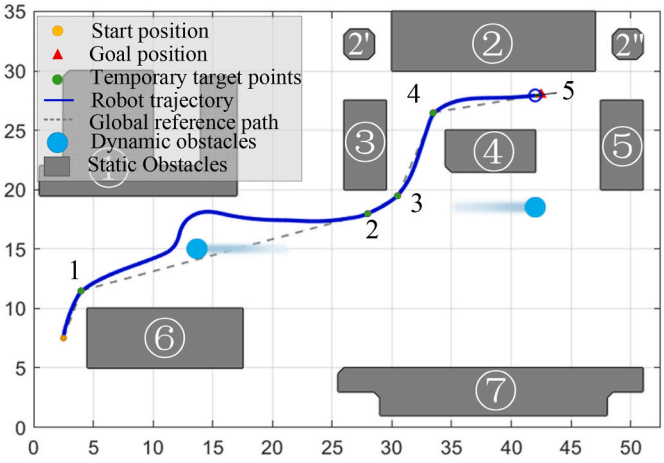


Fig. 8. Experimental results of the proposed A*-DWA. Gray blocks labeled ①–⑦ denote static obstacles. Cyan circles denote dynamic obstacles moving at a constant speed. The yellow point is the start position and the red triangle indicates the final goal. The blue curve represents the robot trajectory. The gray dashed line denotes the global reference path generated by the improved A algorithm. Numbers 1–5 indicate the temporary target points used to guide the local planner.

fully leverage the global optimization potentiality of A* and the local obstacle avoidance advantages of DWA, by that further enhancing the AMR's path planning performance. While reducing path length by 12.54%, the average speed is increased by 13.18%, ensuring the real-time and reliability of the AMR's path planning capabilities. In future research, we will consider multi-objective optimization problems in path planning and the environmental generalization capabilities of algorithms to further improve the intelligent planning capabilities of the AMR.

CRediT authorship contribution statement

Yidao Ji: Writing – original draft, Conceptualization. **Yachen Su:** Software, Resources, Investigation. **Yuxuan Zhao:** Writing – review & editing, Formal analysis. **Cheng Zhou:** Validation, Data curation. **Wei Qian:** Visualization, Investigation. **Wei Wu:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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